NetFix AI

AI-Powered Telecom Intelligence Platform

Graduation Project Thesis

DIGIS GP Team

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A graduation project focused on explainable telecom analytics, engineering decision support, and workflow acceleration.

Project Overview and Business Objective

NetFix AI is an AI-powered telecom engineering platform designed for mobile-network optimization and planning teams. The project addresses a practical operational problem: engineers often work with very large drive-test and planning datasets through fragmented spreadsheets, scripts, plots, and specialist tools. The result is slow investigation, repeated manual validation, inconsistent prioritization, and difficulty turning raw measurements into clear engineering actions.

The platform contains two **independent** engineering workspaces. **Drive Test Intelligence** detects and prioritizes throughput degradation, groups repeated abnormal observations into engineer-facing incidents, and prepares structured evidence for root-cause analysis. **Planning Intelligence** independently supports network-plan visualization, site and sector workflows, conflict detection, and deterministic validation. The two datasets may represent different sites and locations; therefore, planning output is not used to prove or correct a drive-test anomaly, and drive-test observations are not used as ground truth for the planning workflow.

Business objective

The product objective is to reduce the time between "raw telecom data" and an auditable engineering decision. NetFix AI is positioned as B2B engineering software for operator optimization teams, RF planning teams, telecom vendors, managed-service providers, and engineering consultancies. Its value is not to replace the engineer, but to automate repetitive analysis, surface the highest-priority problems, preserve evidence, and make validation and reporting reproducible.

The thesis documents the technical mechanisms behind this objective. The following chapter presents the drive-test throughput anomaly-detection pipeline, including statistical detection, session-safe machine learning, multi-level evidence fusion, episode construction, incident consolidation, and engineering prioritization.

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CHAPTER 1

Drive Test Throughput Anomaly Detection and Prioritization

1.1 Chapter Scope and Role within NetFix AI

This chapter presents the anomaly-detection component of the **Drive Test Intelligence** workstream. The implemented first use case is LTE downlink throughput degradation. The objective is not merely to mark low-throughput samples; it is to identify technically meaningful degradation, distinguish independent evidence from correlated flags, quantify seriousness and confidence, and convert one-second observations into a manageable set of prioritized operational incidents.

The drive-test workflow is deliberately independent from the Planning Intelligence dataset. The planning sites and the drive-test serving cells are not assumed to represent the same physical network area. Consequently, no planning output is used to validate, prove, or correct a drive-test anomaly in this chapter. The anomaly detector reasons only from drive-test measurements, derived features, session-safe predictive references, and the engineering rules defined for the drive-test pipeline.

The implemented detection chain is summarized as

$$\begin{gathered}\text{LTE samples} \rightarrow \text{statistical evidence} \rightarrow \text{ML evidence} \rightarrow \text{row fusion} \\ \rightarrow \text{episodes} \rightarrow \text{operational incidents.}\end{gathered} \tag{1.1}$$

At each stage, the pipeline keeps three concepts separate:

$$\boxed{\text{Impact} \neq \text{Confidence} \neq \text{Priority.}} \tag{1.2}$$

Impact describes the performance loss, confidence describes the strength of supporting evidence, and priority combines the two for investigation ordering.

Engineering design principles

- **Statistical-first detection:** interpretable rules remain the primary anomaly source.
- **Independent evidence families:** correlated low-tail flags are not counted repeatedly as separate confirmation.
- **Session-safe ML:** every production prediction used for anomaly scoring is out-of-fold with respect to the complete drive-test session.
- **Mechanism-aware impact:** radio-limited, resource-limited, capability-gap, and temporal cases are not forced through one inappropriate expected-throughput reference.
- **Conservative aggregation:** isolated one-second spikes are reduced into strict episodes and then compatible operational incidents.
- **Auditable handoff:** raw scores, families, evidence coverage, lineage, and final priority are retained for RCA.

1.2 Data Scope and Modeling Population

The most recent validated execution began with 64,949 raw drive-test rows and 1,365 source columns. Cleaning and feature selection produced a base table of 64,949 rows and 240 columns. The LTE downlink modeling population contained 32,238 one-second observations grouped into 271 drive-test sessions. After pre-anomaly feature engineering, the analysis table contained 32,238 rows and 353 columns.

Artifact / population	Size
Raw drive-test data	64,949 rows × 1,365 columns
Cleaned base table	64,949 rows × 240 columns
LTE-DL modeling population	32,238 rows
Drive-test sessions	271
Pre-anomaly feature table	32,238 rows × 353 columns
Sampling interval used by temporal logic	approximately 1 second

Tab. 1.1: Final-run data scale used by the anomaly-detection pipeline.

The throughput distribution is strongly right-skewed, which is important because symmetric Gaussian assumptions are not suitable for every detector. The median LTE-DL throughput is approximately $38.80\,\mathrm{Mbit\,s^{-1}}$; the empirical global tenth percentile is approximately $4.99\,\mathrm{Mbit\,s^{-1}}$. The global P10 therefore lies very close to the severe low-throughput engineering anchor around $5\,\mathrm{Mbit\,s^{-1}}$, while the broader operational low-throughput reference remains $10\,\mathrm{Mbit\,s^{-1}}$.

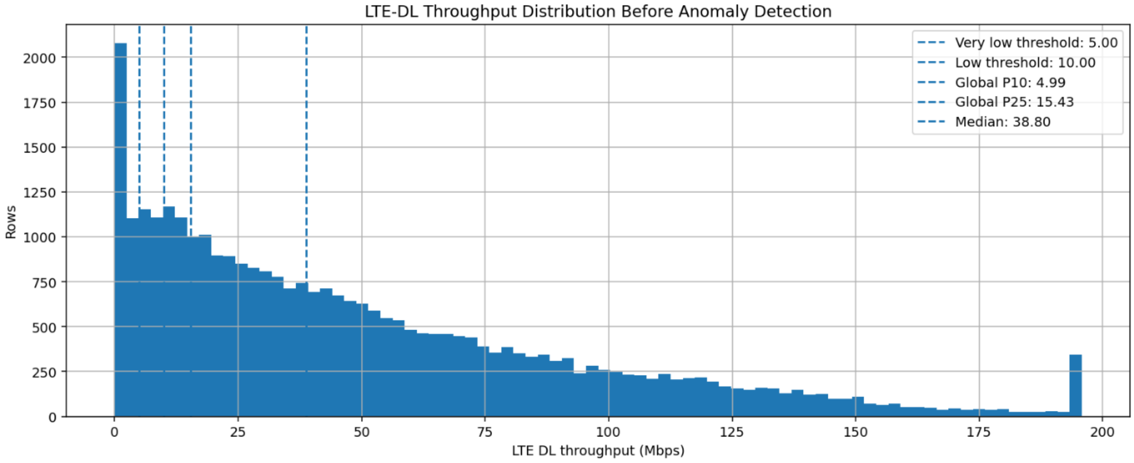

Fig. 1.1: LTE-DL throughput distribution before anomaly detection. The long upper tail motivates robust lower-tail statistics and conditional expected-throughput references rather than one global Gaussian threshold.

1.2.1 Session boundaries and leakage control

Drive-test samples are temporally correlated. A random row split would allow neighboring observations from the same route and session to appear in both model training and validation, artificially improving predictive metrics. The pipeline therefore assigns a session identifier and treats the session as the minimum independence unit for machine-learning validation. Temporal features used by production models are constructed from current or past information only; future samples are not allowed to leak into the prediction of the present row.

1.3 Statistical Detection Layer

The statistical layer intentionally retains multiple views of abnormal behavior because no single threshold can represent all relevant degradation mechanisms. The individual flags are grouped into four independent families for scoring: **low level**, **temporal change**, **expected underperformance**, and **RB efficiency**.

1.3.1 Low-level throughput family

Five complementary low-tail detectors are retained for interpretation.

Fixed engineering threshold

The simplest domain anchor is

$$T_{\text{actual}} < 10\,\mathrm{Mbit\,s^{-1}}. \tag{1.3}$$

It flags 5,549 rows, or 17.21% of the LTE-DL population. This threshold is intentionally interpretable and does not depend on the particular dataset distribution.

Global and session-relative P10

The global empirical tenth percentile is

$$P_{10,\text{global}} = Q_{0.10}(T_{\text{actual}}) = 4.990\,\text{Mbit}\,\text{s}^{-1}, \tag{1.4}$$

which flags 3,224 rows (10.00%). For session s,

$$P_{10,s} = Q_{0.10}(T_{\text{actual}} \mid s), \tag{1.5}$$

and a row is relatively weak when $T_i < P_{10,s(i)}$. The median session threshold is $7.008\,\text{Mbit}\,\text{s}^{-1}$; 2,685 rows (8.33%) satisfy the session-relative rule.

Log-IQR lower fence

Because throughput is positively skewed, the IQR detector works in log space:

$$X = \log(1 + T), \tag{1.6}$$

$$IQR = Q_3(X) - Q_1(X), \tag{1.7}$$

$$L_T = \exp(Q_1(X) - 1.5IQR) - 1. \tag{1.8}$$

The resulting final-run cutoff is approximately $0.669\,\text{Mbit}\,\text{s}^{-1}$, producing 1,279 extreme lower-tail rows (3.97%).

Robust MAD / modified-Z detector

Let $m = \text{median}(X)$ and $MAD = \text{median}(|X - m|)$. The modified robust score is

$$Z_i^* = 0.6745\frac{X_i - m}{MAD}. \tag{1.9}$$

The row is flagged when $Z_i^* \leq -3$. The implied throughput cutoff is approximately $0.550\,\text{Mbit}\,\text{s}^{-1}$, producing 1,145 rows (3.55%). The -3 value is a dimensionless robust-Z threshold, not a dB quantity.

The overlap audit shows an expected nested structure:

$$\text{MAD extreme tail} \subset \text{IQR extreme tail} \subset \text{Global P10}. \tag{1.10}$$

Therefore these flags remain visible in the evidence record, but they do not become three independent votes during fusion.

1.3.2 Temporal-change family

The temporal family detects degradation relative to the row's recent behavior. It includes a rolling/local drop detector and a previous-sample sudden-drop detector. Conceptually, the drop evidence is based on

$$D = \max\left(D_{\text{rolling median}}, D_{\text{previous sample}}\right), \tag{1.11}$$

with continuous drop severity

$$S_{\text{drop}} = \text{clip}\left(\frac{D}{40}, 0, 1\right). \tag{1.12}$$

A $40\,\text{Mbit}\,\text{s}^{-1}$ or larger local collapse saturates this severity term. The independent temporal-change family flags 3,685 rows, or 11.43% of the population.

1.3.3 Expected-underperformance family

Absolute low throughput is not sufficient. A sample can be operationally important even when it is above $10\,\text{Mbit}\,\text{s}^{-1}$ if comparable conditions indicate that much higher throughput should have been achievable. The pipeline therefore constructs conditional P75 opportunity references.

The radio-conditioned reference estimates

$$T_{\text{radio P75}} = Q_{0.75}(T \mid \text{radio context}), \tag{1.13}$$

and the RB-conditioned reference estimates

$$T_{\text{RB P75}} = Q_{0.75}(T \mid \text{resource-demand context}). \tag{1.14}$$

The hierarchical radio P75 table contains 256 reference groups, each satisfying the minimum sample-support rule of 30 rows. The median expected P75 across those groups is approximately $59.99\,\text{Mbit}\,\text{s}^{-1}$. The complete expected-underperformance family flags 1,480 rows (4.59%).

A useful opportunity representation is the expected-minus-actual gap

$$G = T_{\text{expected}} - T_{\text{actual}} \tag{1.15}$$

and the ratio

$$R = \frac{T_{\text{actual}}}{T_{\text{expected}}}. \tag{1.16}$$

These quantities distinguish "low in absolute terms" from "far below a credible context-dependent opportunity."

1.3.4 RB-efficiency family

When resource-block usage is reliably medium or high, throughput per RB provides a complementary efficiency view:

$$\eta_{\text{RB}} = \frac{T_{\text{actual}}}{RB_{\text{used}}}. \tag{1.17}$$

The low-efficiency condition compares η_{RB} with the lower tail among comparable high-demand rows. This family is deliberately narrow: 518 rows, or 1.61% of all LTE-DL observations.

Independent family	Rows	Share
Low level	5,782	17.94%
Temporal change	3,685	11.43%
Expected underperformance	1,480	4.59%
RB efficiency	518	1.61%

Tab. 1.2: Independent statistical families in the final run.

The broader statistical logic produces 8,086 candidate rows (25.08% of the LTE-DL population), of which 6,217 satisfy the statistical RCA-eligibility rules. All configured detector-percentage quality checks passed in the final run; importantly, the narrow expected-throughput and RB-efficiency families did not dominate the candidate population.

1.4 Standalone Statistical Severity Score

The statistical score is not a flag count. It combines a strongest-trigger floor, continuous magnitude, a bounded independent-family bonus, and a bounded context bonus.

1.4.1 Strongest-trigger floor

Each meaningful mechanism supplies a minimum base score, and only the strongest floor is retained:

$$B_{\mathrm{stat}} = \max(B_1, B_2, \ldots, B_k). \tag{1.18}$$

The implemented floors are 15 for any trigger, 25 for low-level throughput, 35 for sustained-low or radio-limited evidence, 50 for temporal drop, 55 for reliable P75 underperformance, and 75 for severe P75 underperformance. Taking a maximum avoids score inflation when multiple correlated flags describe the same physical event.

1.4.2 Continuous magnitude

Four normalized components measure seriousness:

$$S_{\mathrm{ratio}} = \mathrm{clip}\left(\frac{0.60 - R}{0.60}, 0, 1\right), \tag{1.19}$$

$$S_{\mathrm{gap}} = \mathrm{clip}\left(\frac{G}{80}, 0, 1\right), \tag{1.20}$$

$$S_{\mathrm{low}} = \mathrm{clip}\left(\frac{20 - T_{\mathrm{actual}}}{20}, 0, 1\right), \tag{1.21}$$

$$S_{\mathrm{drop}} = \mathrm{clip}\left(\frac{D}{40}, 0, 1\right). \tag{1.22}$$

The combined magnitude is

$$M_{\text{stat}} = 100\left(0.35S_{\text{ratio}} + 0.25S_{\text{gap}} + 0.25S_{\text{low}} + 0.15S_{\text{drop}}\right). \tag{1.23}$$

RB-efficiency evidence can impose an additional magnitude floor when appropriate.

1.4.3 Independent-family and context bonuses

If N_f independent statistical families support the row,

$$B_{\text{family}} = \text{clip}\left(5\ln\left[1 + \max(N_f - 1, 0)\right], 0, 12\right). \tag{1.24}$$

Thus one family contributes no bonus, while four independent families contribute only $5\ln 4 \approx 6.93$ points. The context bonus is also bounded:

$$B_{\text{context}} = \text{clip}\left(3L + 4H_{\text{RB}} + 2C_{\text{CA}} + 2E_{\text{event}} + 2R_{\text{poor radio}}, 0, 10\right). \tag{1.25}$$

The pre-compression score is

$$S_{\text{stat,pre}} = \max(B_{\text{stat}}, M_{\text{stat}}) + B_{\text{family}} + B_{\text{context}}. \tag{1.26}$$

Low-RB observations without strong radio or temporal explanation are smoothly compressed toward a lower ceiling because low throughput under low resource demand can represent low offered load rather than network limitation. Non-catastrophic scores above 85 are also softly compressed toward 95, preserving ranking without creating an artificial score pile-up. A value of 100 is reserved for a strict catastrophic override requiring very low actual throughput, a high expected reference, credible medium/high RB demand, and support from at least two independent statistical families.

The contribution audit confirms that the score is driven mainly by anomaly seriousness rather than bonuses. Among final statistical anomaly rows, the mean strongest-trigger base was 44.47 and the mean continuous magnitude was 55.74, whereas the mean family and context bonuses were only 1.31 and 3.85 respectively. Candidate-score quality control also showed a median score of 58.70, P90 of 87.55, P95 of 92.90, and no artificial pile-up at either compression boundary.

1.5 Session-Safe Machine-Learning Evidence

Machine learning is a supporting evidence source rather than the primary detector. Three models are allowed to participate in the production decision:

1. HistGradientBoosting strict-causal temporal central predictor;
2. HistGradientBoosting resource-conditioned central predictor;
3. calibrated resource-conditioned Q75 predictor.

A sequence LSTM-Q50 model and Isolation Forest remain validation/context methods and do not become additional production evidence families.

1.5.1 Group-aware out-of-fold prediction

For row i belonging to session $g(i)$, the prediction is generated by a model trained without that complete session:

$$\hat{T}_i = f_{-g(i)}(X_i). \tag{1.27}$$

The final run used five group-aware folds, so every one of the 32,238 rows receives an out-of-fold prediction that cannot directly memorize its own session.

Model	**Train R^2**	**Held-out R^2**	**Train MAE**	**Held-out MAE**
HGB strict-causal temporal	0.945	0.920	6.93	8.50
HGB resource-conditioned	0.936	0.899	7.63	9.76
Calibrated resource Q75	0.836	0.809	12.42	13.75
LSTM-Q50 validation comparator	0.807	0.749	13.31	14.74

Tab. 1.3: Predictive-model quality from the final validation run.

The temporal HGB model is the strongest central predictor. The Q75 model has a different role: it approximates an upper achievable opportunity reference and is therefore not expected to minimize central-prediction error.

1.5.2 Per-model anomaly evidence

For model m,

$$G_m = \hat{T}_m - T_{\text{actual}}, \tag{1.28}$$

$$R_m = \frac{T_{\text{actual}}}{\hat{T}_m}, \tag{1.29}$$

$$e_m = \log(1 + T_{\text{actual}}) - \log(1 + \hat{T}_m). \tag{1.30}$$

The magnitude gate requires an expected value of at least $25\,\mathrm{Mbit\,s^{-1}}$, a gap of at least $15\,\mathrm{Mbit\,s^{-1}}$, and a sufficiently poor actual/expected ratio. The resource Q75 and resource central models use a ratio gate of 0.40; the temporal central model uses 0.50. When RB usage is reliable, credible medium/high RB demand is also required.

Residual evidence must also fall in the model's out-of-fold lower tail. Continuous evidence combines ratio, gap, and residual-tail severity:

$$S_{\mathrm{ratio},m} = \mathrm{clip}\left(\frac{r_m - R_m}{r_m}, 0, 1\right), \tag{1.31}$$

$$S_{\mathrm{gap},m} = \mathrm{clip}\left(\frac{G_m - 15}{65}, 0, 1\right), \tag{1.32}$$

$$S_{\mathrm{tail},m} = \mathrm{clip}\left(\frac{0.10 - p_m}{0.10}, 0, 1\right), \tag{1.33}$$

$$E_m = 0.35 S_{\mathrm{ratio},m} + 0.35 S_{\mathrm{gap},m} + 0.30 S_{\mathrm{tail},m}. \tag{1.34}$$

Reliability priors are 0.95 for the temporal HGB and 0.90 for both resource models. The two correlated resource models are merged by maximum,

$$E_{\mathrm{resource}} = \max(0.90 E_{Q75}, 0.90 E_{\mathrm{HGB,resource}}), \tag{1.35}$$

while the temporal family is

$$E_{\mathrm{temporal}} = 0.95 E_{\mathrm{HGB,temporal}}. \tag{1.36}$$

The two independent production families are then combined using noisy-OR:

$$C_{\mathrm{ML,base}} = 1 - (1 - E_{\mathrm{resource}})(1 - E_{\mathrm{temporal}}). \tag{1.37}$$

Only 322 rows (0.999%) reached the ML candidate gate, 140 rows (0.434%) became final strict production-ML anomalies, and only four rows (0.012%) were ultimately retained through an ML-only supervised exception path. All four require human review. This confirms that ML is intentionally selective rather than a broad replacement for the statistical layer.

1.6 Row-Level Statistical–ML Fusion

The final row-level fusion does not average statistical and ML scores. Instead, it derives bounded evidence confidence and a mechanism-aware impact score.

1.6.1 Evidence confidence

Statistical evidence uses the uncapped analytical score with a mechanism reliability prior:

$$E_{\mathrm{stat}} = \mathrm{clip}(S_{\mathrm{stat,raw}}, 0, 1) R_{\mathrm{stat}}. \tag{1.38}$$

The reliability prior is highest for reliable P75, RB-efficiency, or strict CA evidence, and lower for low-tail-only cases. Production ML evidence is

$$E_{\mathrm{ML}} = 1 - (1 - E_{\mathrm{resource}})(1 - E_{\mathrm{temporal}}). \tag{1.39}$$

The base row confidence is

$$C_{\text{base}} = 1 - (1 - 0.90E_{\text{stat}})(1 - 0.85E_{\text{ML}}), \tag{1.40}$$

followed by bounded agreement and demand bonuses:

$$C_{\text{row}} = \text{clip}\left(C_{\text{base}} + 0.07A + 0.03R + 0.03M_2, 0, 1\right). \tag{1.41}$$

Here A denotes statistical–ML agreement, R indicates medium/high RB demand, and M_2 indicates support from both production ML families. The reported row confidence is $100C_{\text{row}}$.

1.6.2 Mechanism-aware impact and priority

Rather than taking the largest of unrelated expected-throughput predictions, the system builds mechanism-specific references for capability, resource, and temporal behavior. A separate radio-limited impact formulation avoids comparing poor-radio rows with an unrealistic high-capability expectation. The final row impact is

$$I_{\text{row}} = 100 \max\left(I_{\text{capability}}, I_{\text{resource}}, I_{\text{temporal}}, I_{\text{radio}}\right). \tag{1.42}$$

The row priority is

$$\boxed{Priority_{\text{row}} = 0.55I_{\text{row}} + 0.45Confidence_{\text{row}}.} \tag{1.43}$$

Impact receives a slight majority weight so a severe degradation remains important, while confidence can still materially change investigation order.

The final row-level handoff contains 3,574 observations: 3,434 statistical-only retained rows (96.08%), 136 statistical–ML consensus rows (3.81%), and four supervised ML-only rows (0.11%). The dominant row-impact families were radio (1,421), capability (1,115), temporal (973), and resource (65).

The selected impact–confidence fusion also outperformed naive score averaging as an operational ranking mechanism. Among final handoff rows, the selected method produced a median priority of 68.31 and P90 of 83.25, compared with a median of 35.13 for a naive statistical/ML average. This matters because a missing ML score should not suppress a strong interpretable statistical anomaly.

1.7 Episode Construction and Persistence

One-second rows are too granular for direct RCA. Final fused rows are grouped within the same session when timestamp gaps are at most three seconds and serving-cell continuity is compatible. This creates candidate episodes that represent continuous or near-continuous degradation.

Fig. 1.2: Distribution of final fused row-level priority. The row score represents the seriousness and evidence strength of an exact one-second observation before temporal aggregation.

For an episode, longest uninterrupted run, anomalous time, and density are normalized as

$$S_{\text{run}} = \text{clip}\left(\frac{L_{\text{run}}}{10}, 0, 1\right), \tag{1.44}$$

$$S_{\text{time}} = \text{clip}\left(\frac{T_{\text{anomalous}}}{20}, 0, 1\right), \tag{1.45}$$

$$Density = \frac{T_{\text{anomalous}}}{T_{\text{inclusive span}}}. \tag{1.46}$$

Episode persistence is

$$P_{\text{episode}} = 100\left(0.60S_{\text{run}} + 0.25S_{\text{time}} + 0.15Density\right). \tag{1.47}$$

Impact and confidence are aggregated as

$$I_{\text{episode}} = 0.50I_{\text{row,max}} + 0.25I_{\text{row,mean}} + 0.25P_{\text{episode}}, \tag{1.48}$$

$$C_{\text{episode}} = 0.50C_{\text{row,max}} + 0.25C_{\text{row,mean}} + 0.15A_{\text{episode}} + 0.10E_{\text{episode}}, \tag{1.49}$$

where A_{episode} is statistical–ML agreement coverage and E_{episode} is high-quality direct-evidence coverage. The episode priority remains

$$Priority_{\text{episode}} = 0.55I_{\text{episode}} + 0.45C_{\text{episode}}. \tag{1.50}$$

The final run formed 2,087 candidate fused episodes. Strict gating retained 1,429 final episodes containing 2,824 fused anomaly rows. Of the retained episodes, 1,362 passed the normal priority-and-confidence gate and 67 passed a verified critical-evidence override. Median episode persistence was 22.25.

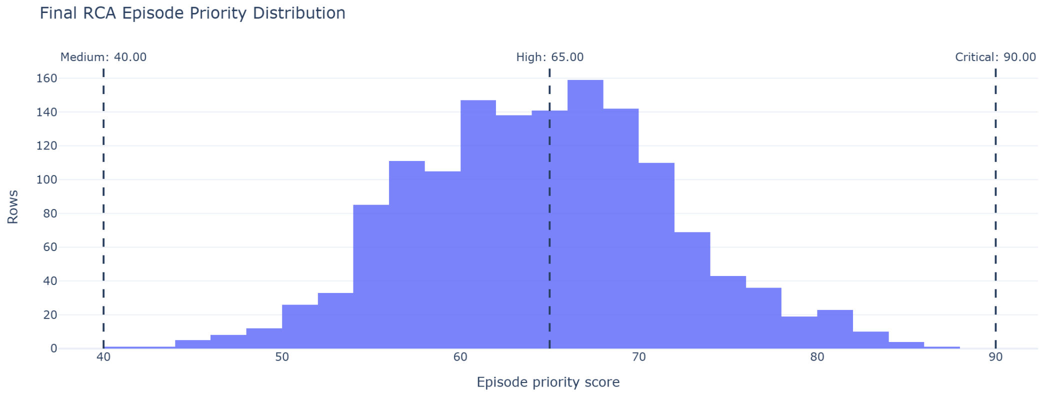

Fig. 1.3: Priority distribution of strict final episodes. Episode scoring broadens the question from "how bad was this second?" to "how severe, persistent, and well-supported was the complete anomaly period?"

1.8 Operational Incident Consolidation

Strict episodes can still represent the same engineer-facing network problem when separated by a short recovery or internal gap. Compatible episodes are therefore consolidated when the temporal gap is at most five seconds, spatial distance is at most approximately 50 m, the session is the same, the serving-cell relationship is compatible, and the dominant mechanism is compatible.

Incident persistence introduces recurrence explicitly:

$$S_{\text{incident run}} = \operatorname{clip}\left(\frac{L_{\max}}{10}, 0, 1\right), \tag{1.51}$$

$$S_{\text{incident time}} = \operatorname{clip}\left(\frac{T_{\text{total anomalous}}}{20}, 0, 1\right), \tag{1.52}$$

$$S_{\text{recurrence}} = \operatorname{clip}\left(\frac{N_{\text{strict episodes}}}{3}, 0, 1\right), \tag{1.53}$$

$$P_{\text{incident}} = 100\left(0.50 S_{\text{incident run}} + 0.30 S_{\text{incident time}} + 0.20 S_{\text{recurrence}}\right). \tag{1.54}$$

Incident impact and confidence are

$$I_{\text{incident}} = 0.45 I_{\text{episode,max}} + 0.30 I_{\text{episode,mean}} + 0.25 P_{\text{incident}}, \tag{1.55}$$

$$C_{\text{incident}} = 0.50 C_{\text{episode,max}} + 0.30 C_{\text{episode,mean}} + 0.10 A_{\text{incident}} + 0.10 E_{\text{incident}}. \tag{1.56}$$

The canonical engineer-facing priority is

$$\boxed{Priority_{\text{incident}} = 0.55 I_{\text{incident}} + 0.45 C_{\text{incident}}.} \tag{1.57}$$

In the final payload this value is stored as `incident_final_priority_0_100`.

Only 47 episode-to-episode bridges were accepted, showing that consolidation is conservative. The 1,429 strict episodes became 1,382 operational incidents. Most incidents contain exactly one strict episode; only 38 contain multiple episodes. The final incident-priority range is 34.31–78.41 with a median of 56.51. Severity counts are 11 Low, 1,224 Medium, and 147 High; no incident reaches the Critical threshold.

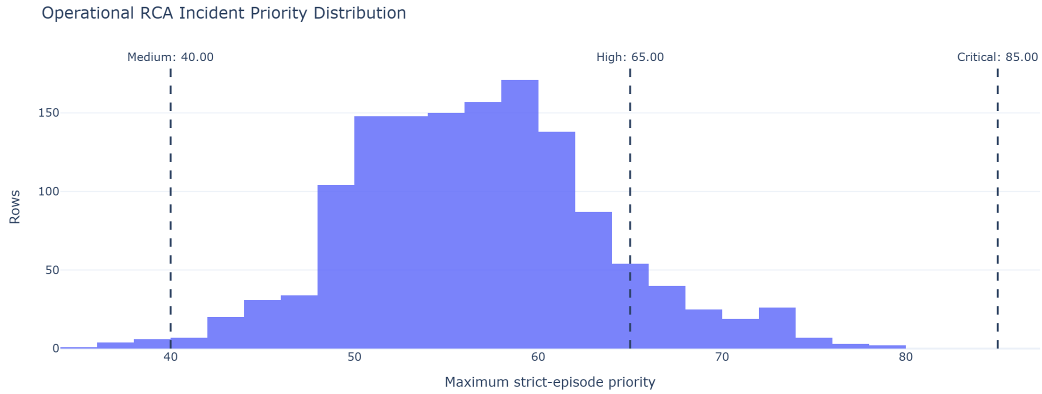

Fig. 1.4: Operational RCA incident-priority distribution. The incident score is intentionally more conservative than the peak row score because it represents the complete engineer-facing problem after persistence, recurrence, and evidence consistency are considered.

1.9 End-to-End Reduction and Final Handoff

The final detection funnel is

$$32{,}238 \rightarrow 8{,}086 \rightarrow 6{,}217 \rightarrow 140 \rightarrow 3{,}574 \rightarrow 1{,}429 \rightarrow 1{,}382. \tag{1.58}$$

This reduction is controlled consolidation rather than data loss. Broad candidates are filtered by mechanism quality; correlated statistics are grouped; ML is required to meet strict predictive-disagreement conditions; weak isolated rows are removed; continuous observations become episodes; and nearby compatible episodes become incidents.

The compact handoff contains 1,382 unique incident IDs and matching canonical priorities across the operational CSV, compact CSV, and compact JSON artifacts. This is important for downstream RCA because every engineer-facing problem has one stable identifier and one canonical ranking score, while peak-row and episode priorities remain available as supporting forensic context.

1.10 Representative Detected Degradation

Figure 1.6 shows a high-priority final episode around the selected anomaly instant. The figure overlays actual throughput, statistical opportunity references, resource and temporal model predictions, and the rolling baseline. The selected episode has priority approximately 87.0, impact 76.4, confidence 100, and persistence 22.2. At the anomaly instant the actual throughput is approximately $3.18\,\mathrm{Mbit\,s^{-1}}$, while the radio-conditioned P75

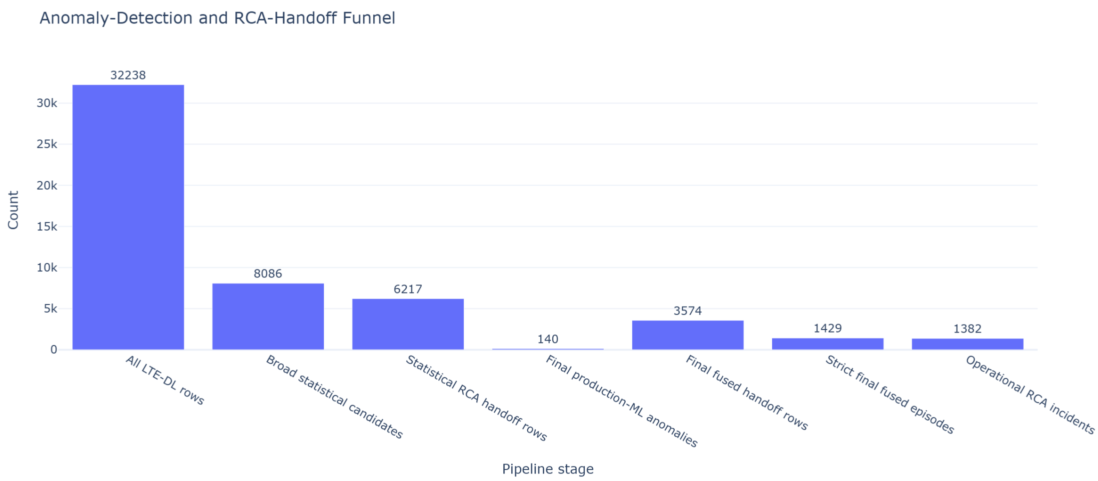

Fig. 1.5: Final anomaly-detection and RCA-handoff funnel for the most recent run. Statistical and ML paths are not additive counts at every intermediate stage; they converge at row-level fusion before episode and incident aggregation.

opportunity reference is approximately 118.45 $\mathrm{Mbit\,s^{-1}}$ and the RB-conditioned P75 reference is approximately 107.87 $\mathrm{Mbit\,s^{-1}}$.

The important point is not simply that throughput is low. Multiple independent views agree that the row is far below a credible opportunity: the statistical P75 references are high, the production predictive families also expect substantially higher throughput, and the local temporal context indicates a sharp collapse. This produces high confidence without relying on a single threshold.

The same methodology also detects cases that fixed low-throughput thresholds alone would miss. For example, a sudden degradation can remain above 10 $\mathrm{Mbit\,s^{-1}}$ but still show a very large local drop or expected-throughput gap. Conversely, radio-limited cases are preserved even when an expected-capability comparison would be inappropriate; they use a radio-specific impact formulation based on absolute low throughput and persistence instead of exaggerating the gap to an unrelated high prediction.

1.11 Serving-Cell Prioritization Derived from Detected Behavior

After row, episode, and incident construction, the notebook also aggregates behavior by serving-cell identity to provide an engineering prioritization view. This ranking is not used to decide whether an individual row is anomalous; it is a downstream summary of cell performance across the available observations.

Each cell receives data-driven percentile badness values. Low values are bad for median throughput, P10 throughput, CQI, MCS, SINR, RSRQ, and RSRP; high values are bad for anomaly rate, P75 opportunity gap, and BLER. The final score is a weighted average

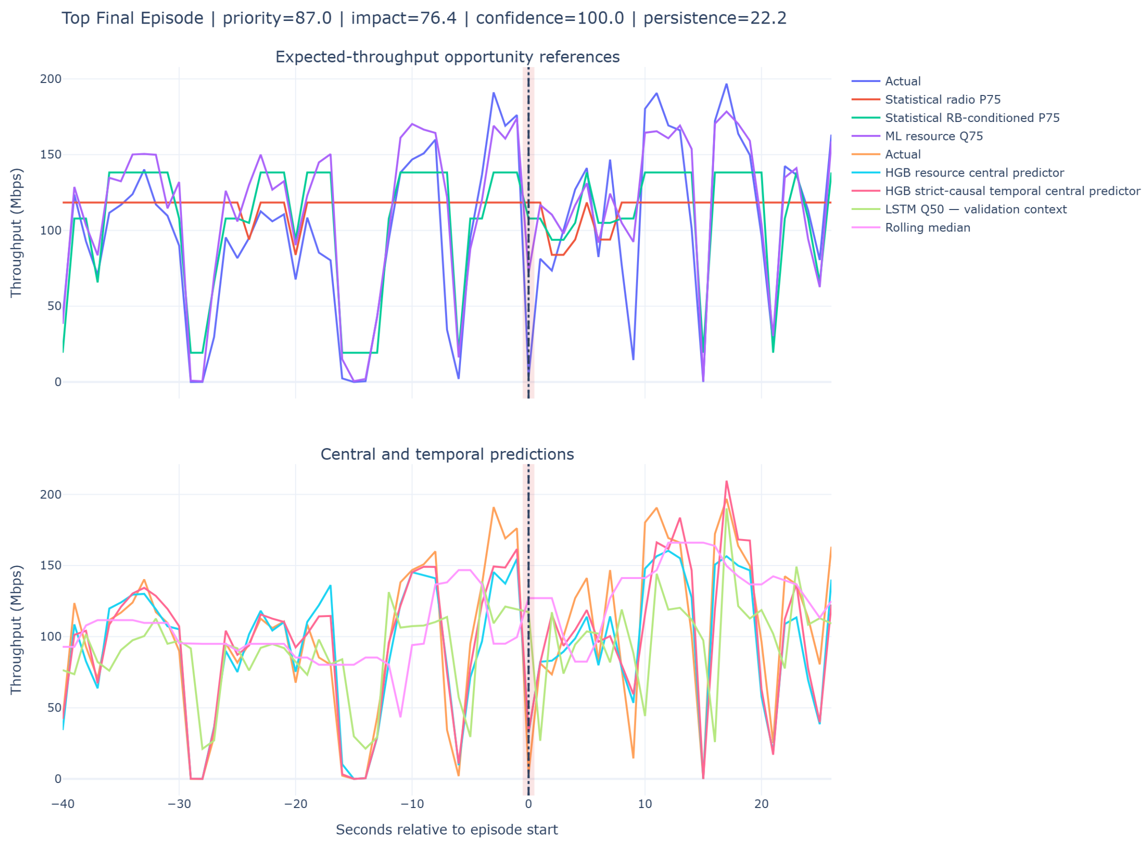

Fig. 1.6: Representative high-priority fused episode. The shaded region marks the selected anomaly period; the upper panel emphasizes opportunity references and the lower panel shows central/temporal predictive context.

of the available percentile badness terms:

$$Score_{\mathrm{ECI}} = 100\frac{1}{\sum w_j}\Big(0.25B_{\mathrm{medTP}} + 0.12B_{\mathrm{P10TP}} + 0.13B_{\mathrm{anom}} + 0.10B_{\mathrm{P75gap}} \tag{1.59}$$

$$+ 0.10B_{\mathrm{CQI}} + 0.08B_{\mathrm{MCS}} + 0.10B_{\mathrm{BLER}} + 0.06B_{\mathrm{SINR}} \tag{1.60}$$

$$+ 0.04B_{\mathrm{RSRQ}} + 0.02B_{\mathrm{RSRP}}\Big). \tag{1.61}$$

Weights are renormalized over metrics actually available for the entity. Therefore, the ranking is multi-KPI and percentile-based rather than a simple lowest-throughput list.

In the final run, ECI 1079182 is ranked first with a bad-cell score of 90.37/100. It has 35 observations across two sessions, median throughput of 7.28 Mbit s^{-1}, P10 throughput of 1.96 Mbit s^{-1}, statistical anomaly rate of 91.43%, and median P75 opportunity gap of 37.18 Mbit s^{-1}. Its ranking reliability is labeled Low because the supporting sample and session counts are limited; the score therefore prioritizes investigation without pretending that evidence volume is high.

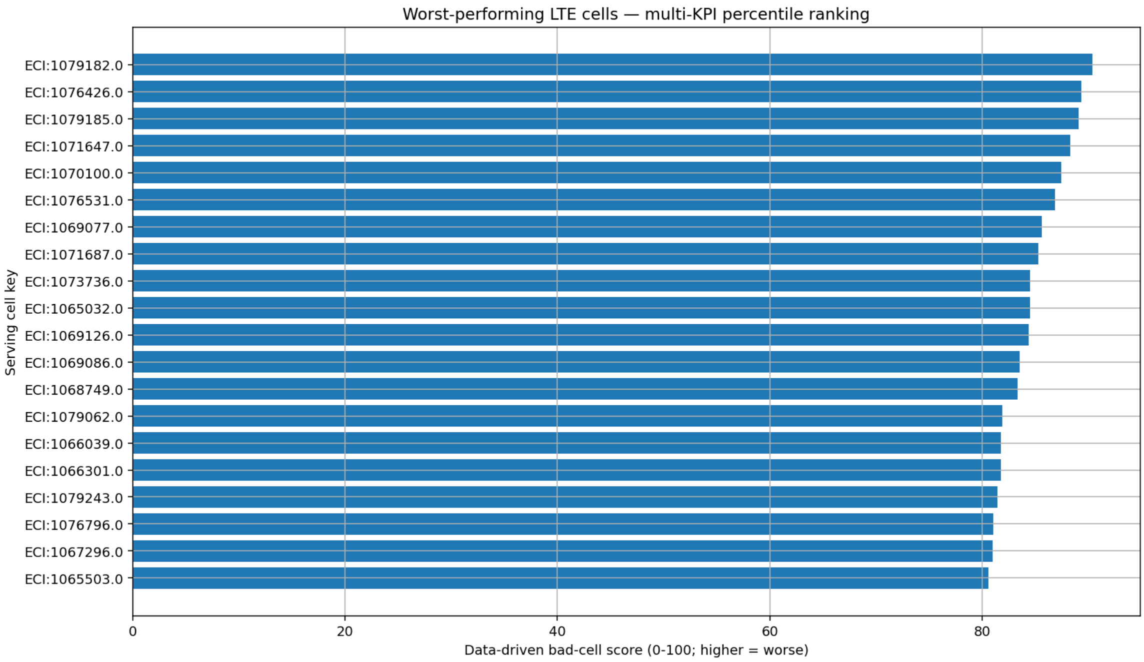

Fig. 1.7: Worst-performing LTE serving cells ranked by the multi-KPI percentile badness score. Ranking reliability must be interpreted together with the score.

1.12 Validation, Governance, and Limitations

The final pipeline contains several explicit safeguards.

- **No planning-data leakage:** Planning Intelligence is an independent workstream and does not validate drive-test anomalies.
- **No same-session ML leakage:** production ML evidence is generated out-of-fold by complete session groups.

- **Causal temporal features:** features intended for production temporal prediction use past/current information only.
- **Correlated methods are grouped:** P10, IQR, and MAD remain auditable but do not count as three independent families.
- **Validation-only models remain validation-only:** LSTM and Isolation Forest do not silently become production votes.
- **ML-only cases are supervised:** the four retained ML-only rows require human review.
- **Expected throughput is an opportunity reference:** it is not a guaranteed physical capacity value and should not be interpreted as such.
- **Location is an observation location:** incident centroids describe where the anomaly was observed, not verified tower coordinates. Most final incident locations have limited location-reliability evidence, so map displays must use careful wording.
- **Cell rank and rank reliability are separate:** a high bad-cell score based on few rows should trigger investigation, not an automatic network change.

Final-run integrity

The most recent execution produced 3,574 final fused rows, 1,429 strict episodes, and 1,382 operational incidents. RCA handoff integrity checks passed, incident identifiers were unique, the canonical priority agreed across operational and compact artifacts, and the governed output reused the validated out-of-fold prediction artifacts rather than silently retraining models during handoff generation.

1.13 Chapter Summary

The implemented anomaly detector converts a large one-second LTE drive-test table into a small, auditable set of operational problems. Interpretable statistical methods remain primary, while session-safe ML contributes only when predictive disagreement is strong. Correlated evidence is grouped before confidence fusion, performance loss is modeled through mechanism-aware impact, and the final priority is computed consistently as 55% impact and 45% confidence. Temporal aggregation then changes the meaning of the score from instantaneous severity to episode seriousness and finally to operational investigation priority.

The resulting handoff contains 1,382 engineer-facing incidents rather than thousands of disconnected low-throughput samples. Each record preserves the evidence required by the next RCA stage: affected time/session/cell context, anomaly mechanism, supporting KPI values, impact, confidence, persistence, priority, and lineage back to the underlying observations. This provides the structured and governed input used by the root-cause-analysis chapter without claiming any dependency on the separate network-planning dataset.

CHAPTER 2

Root Cause Analysis

CHAPTER 3

Planning Intelligence

3.1 Overview and Scope

Planning Intelligence is the second, independent workspace of NetFix AI. Where the Drive Test Intelligence workspace (Chapter 1) turns raw throughput measurements into prioritized incidents, Planning Intelligence solves a different engineering problem: assigning radio identity parameters to every sector of a live 5G NR network so that neighboring cells cannot be confused with one another, while respecting the physical and combinatorial constraints imposed by 3GPP identity ranges.

Scope of this chapter

This chapter documents the NR5G PCI/RSI planning engine end to end: the seven-stage pipeline, the deterministic assignment algorithms, the independent validator, and the measured results of running the engine on a live Egyptian Band N41 deployment spanning **2,554 sectors** across **725 sites** in four regions — Alexandria (`ALX`), Sinai (`SIN`), Upper Egypt (`UPP`), and Delta (`DEL`).

As noted in **??** of the project overview, the planning dataset and the drive-test dataset are independent: planning output is never used to prove or correct a drive-test anomaly, and drive-test observations are never treated as ground truth for planning decisions.

3.1.1 The Three Assigned Parameters

Every sector requires three interrelated identity values, summarized in Table 3.1.

Parameter	3GPP range	Role
PCI (Physical Cell Identity)	0–1007	Primary identity broadcast by the cell; must not collide or be confused with nearby cells. Defined as $N_{ID} = 3N_{ID}^{(2)} + N_{ID}^{(1)}$.
RSI (Root Sequence Index)	0–830	Determines the PRACH root sequence set; must not repeat within the reuse distance or random-access preambles collide.
Mod4 = PCI mod 4	0–3	Groups PCIs for SRS/DMRS orthogonality; four possible groups drive inter-site interference planning.
Mod3 = PCI mod 3	0–2	Groups PCIs for PSS/SSS synchronization-signal separation; enforced ring-adjacently on co-site sectors.

Tab. 3.1: Radio identity parameters assigned by the planning engine

Mod3 and Mod4 are not independently assigned — they are algebraic consequences of the chosen PCI. However, the engine plans Mod4 *before* PCI (Section 3.4.1) and then searches for a PCI that both satisfies the hard identity rules and lands on the already-chosen Mod4 group, since Mod4 has a harder combinatorial ceiling (only 4 values) than Mod3 (3 values) against the same neighbor density.

3.1.2 Architecture Pivot: Full Reset vs. Incremental Planning

Two architectural strategies were evaluated for re-planning the network:

- **Incremental (Option 2):** preserve every already-assigned PCI/RSI and patch only sectors that are flagged as requiring planning.
- **Full reset (Option 1):** discard every existing assignment and re-plan all 2,554 sectors from scratch under one consistent rule set.

Option 1 was chosen. A full reset removes path-dependent legacy artifacts (assignments made under earlier, looser rules) and guarantees that the final plan is provably consistent end to end, rather than consistent only with respect to whichever subset happened to be replanned. All results in this chapter are from a full-reset run.

3.2 System Architecture

The engine is implemented as nine flat, function-based Python modules — no classes, no dataclasses, plain dictionaries throughout the data flow. Each module has exactly one responsibility, summarized in Table 3.2.

Stage	**Module**	**Input ⇒ Output**
1. Load	`loader.py`	Excel workbook ⇒ validated, sorted `list[dict]` of sector records
2. Neighbor graph	`neighbor_graph.py`	sectors ⇒ tiered neighbor graph (7 km / 14 km)
3. Geometry	`facing.py`	sector pair ⇒ alignment / interference potential
4. Constraints	`constraints.py`	sector + graph + assignments ⇒ forbidden PCI/RSI/Mod3 sets
5. Assignment	`assignment_engine.py`	sectors + graph ⇒ `{site_sector_id: {pci, rsi, mod4}}`
6. Validation	`validator.py`	sectors + graph + assignments ⇒ independent KPI report
7. Export	`exporter.py`	assignments + report ⇒ workbook with `KPI_Report` sheet

Tab. 3.2: Pipeline modules and their responsibilities

Two orchestration modules sit above the shared engine (`assignment_engine.py`): `bulk_plan.py` plans the whole network region by region, and `add_site.py` plans a single new site against a region that has already been planned. Both call the identical `assign_all()` function; the only difference is what is pre-populated in `existing_assignments`. Figure 3.1 shows the full data flow.

3.2.1 Why Regions Can Be Planned Independently

`bulk_plan.py` splits the network into four regions using the 3-letter site-ID prefix (`region_of()`). This is only correct because analysis of the real network confirmed that **0% of tier-1 (7 km) and tier-2 (14 km) neighbor relationships cross a region boundary**. Coloring each region's subgraph independently therefore gives exactly the same result as one global coloring, at a fraction of the computational cost — regions are processed largest-first so that the most constrained (densest) subgraphs are solved while the most wall-clock budget is still available.

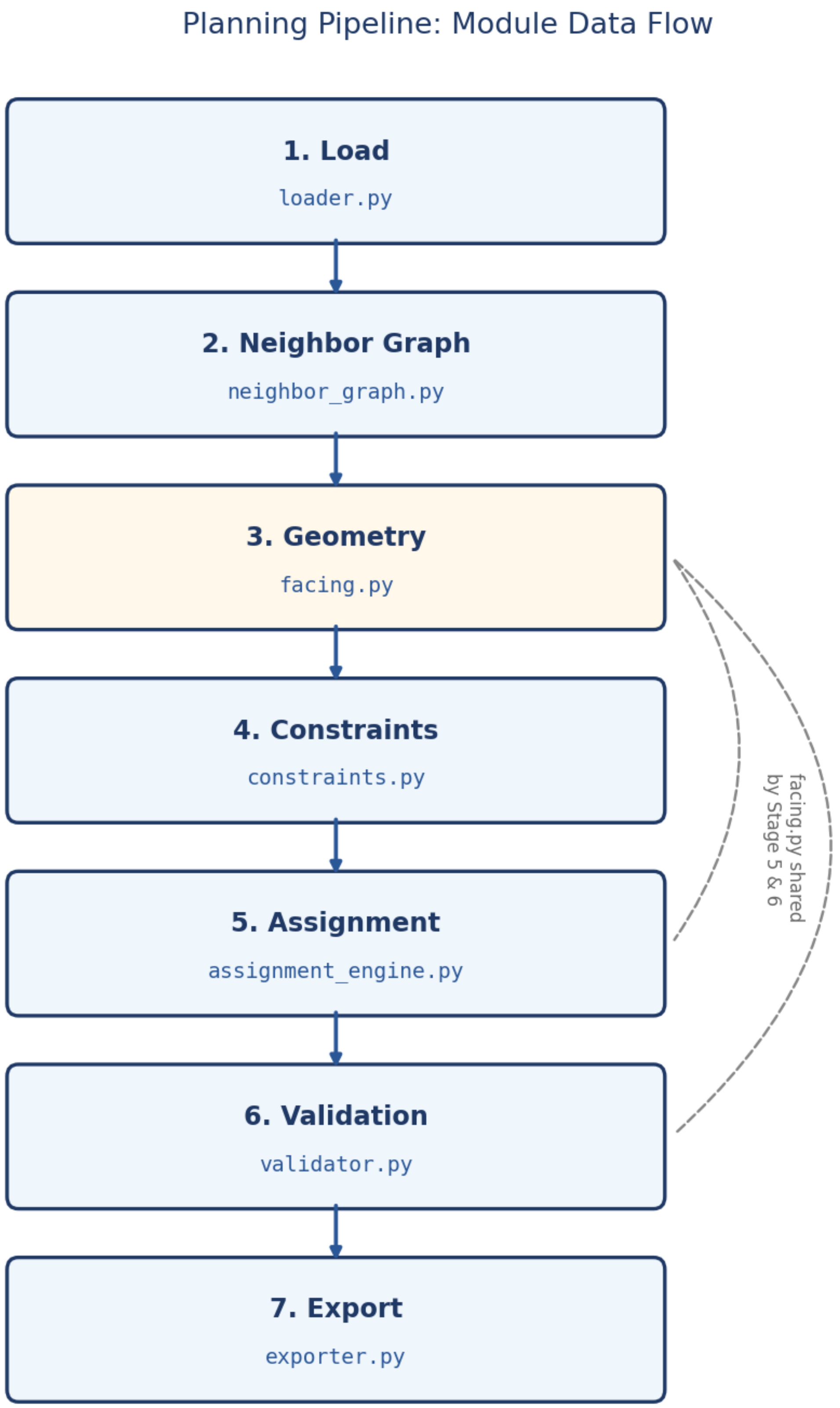

Fig. 3.1: End-to-end planning pipeline. Geometry (Stage 3) is shared between the assignment engine (Stage 5) and the independent validator (Stage 6), guaranteeing both agree on what “facing” means.

3.3 Pipeline Stages in Detail

3.3.1 Stage 1: Data Loading

`load_sectors()` reads the `Planning` sheet with `openpyxl`, validates every row, and returns sectors sorted deterministically by `(site_id, sector_id)`. Validation checks latitude/longitude bounds, azimuth $\in [0, 360)$, PCI/RSI range membership, PCI/Mod4 arithmetic consistency, and duplicate `site_sector_ID` detection — every problem found is reported at once rather than failing on the first error. A `plan_all` flag makes every unlocked sector a planning target regardless of the sheet's `Requires_Planning` column; this is what the full-reset architecture uses instead of the sheet-flagged incremental mode.

Field	**Description**
`site_sector_id`	Unique key, e.g. `ALX2014-1`
`site_id`	Site identifier; first 3 characters encode the region
`sectors_count`	Number of co-located sectors on this site (drives site-size-aware Mod3/Mod4 rules)
`azimuth`	Antenna pointing direction in degrees
`lat`, `lon`	Geographic coordinates
`rsi`, `pci`, `mod4`	`None` if unassigned; a non-`None` PCI marks the sector as *locked*

Tab. 3.3: Sector record structure (loader.py)

Measured on the raw dataset: 2,554 sectors across 725 sites, all requiring planning under the full-reset flag, single band `N41`.

3.3.2 Stage 2: Neighbor Graph Construction

Latitude/longitude is projected onto a local flat-earth East-North-Up (ENU) plane centered on the dataset centroid — accurate enough for cell-planning distances (tens of kilometers) without a full geodesy library. A KD-tree (`scipy.spatial.cKDTree`) then answers two radius queries:

- **Tier-1 (direct, 7 km):** sectors that can physically see each other — must not share PCI or RSI (collision).
- **Tier-2 (extended, 14 km):** sectors within this wider radius must not share PCI either (confusion), since a UE could report the same PCI from two genuinely different physical cells; RSI reuse is also checked at this radius.

Neighbor lists are sorted by distance, and the whole graph is cached to disk keyed on a hash of sector IDs, coordinates, and the two radii, so re-running on an unchanged dataset skips the KD-tree rebuild entirely. Sectors co-located on the same site are always mutual tier-1 neighbors at distance zero — the tightest possible reuse-distance case, and intentional.

Metric	Value
Nodes	2,554
Tier-1 radius	7,000 m
Tier-2 radius	14,000 m
Average tier-1 neighbors per sector	99.84
Maximum tier-1 neighbors	280
Average tier-2 neighbors per sector	150.48
Maximum tier-2 neighbors	344
Isolated nodes (no tier-1 neighbor)	0

Tab. 3.4: Measured neighbor graph statistics, full dataset

Structural implication

An average of ∼100 tier-1 neighbors competing for only 4 Mod4 values is the binding density constraint referenced throughout Section 3.4.1 and Section 3.9: it is the reason a nonzero Mod4 inter-site clash count is structurally expected, not a defect in the assignment algorithm.

3.3.3 Stage 3: Geometry Engine

`facing.py` is shared verbatim by the assignment engine (Stage 5) and the validator (Stage 6), which guarantees the planner and the independent auditor can never disagree about what "facing" means. Every function is parameter-free: no tuned beamwidth cutoff or arbitrary threshold constant is used anywhere in the geometry model.

$$\texttt{bearing\_deg}(A,B) = \text{great-circle compass bearing from } A \text{ to } B \tag{3.1}$$

$$\texttt{angle\_diff}(a,b) = \min(|a-b| \bmod 360,\ 360-(|a-b| \bmod 360)) \tag{3.2}$$

$$\texttt{distance\_m}(A,B) = \text{haversine great-circle distance} \tag{3.3}$$

$$\texttt{alignment}(A,B) = \max\Big(0,\ \cos(\texttt{angle\_diff}(\text{azimuth}_A,\ \texttt{bearing\_deg}(A,B)))\Big) \tag{3.4}$$

$$\texttt{interference\_potential}(A,B) = \frac{\texttt{alignment}(A,B)+\texttt{alignment}(B,A)}{\texttt{distance\_m}(A,B)^2} \tag{3.5}$$

Equation (3.4) returns 0 for co-located sectors (bearing undefined) or angular offsets $\geq 90°$, and 1 for a dead-ahead offset of $0°$. Figure 3.2 plots this relationship.

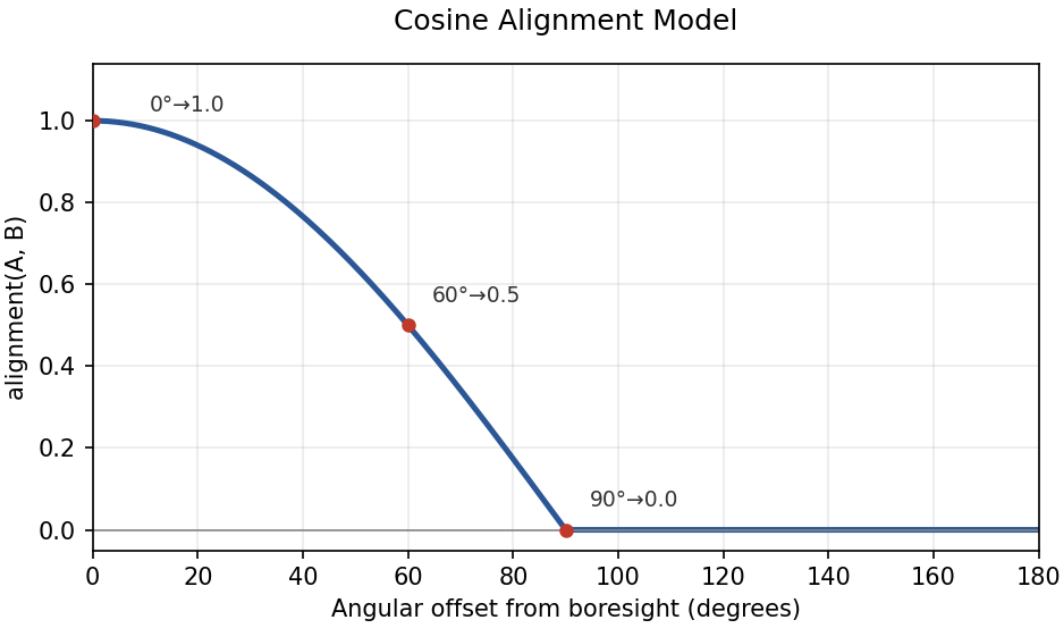

Fig. 3.2: Cosine alignment model: alignment vs. angular offset from boresight. Key points: $0° \to 1.0$, $60° \to 0.5$, $90° \to 0.0$.

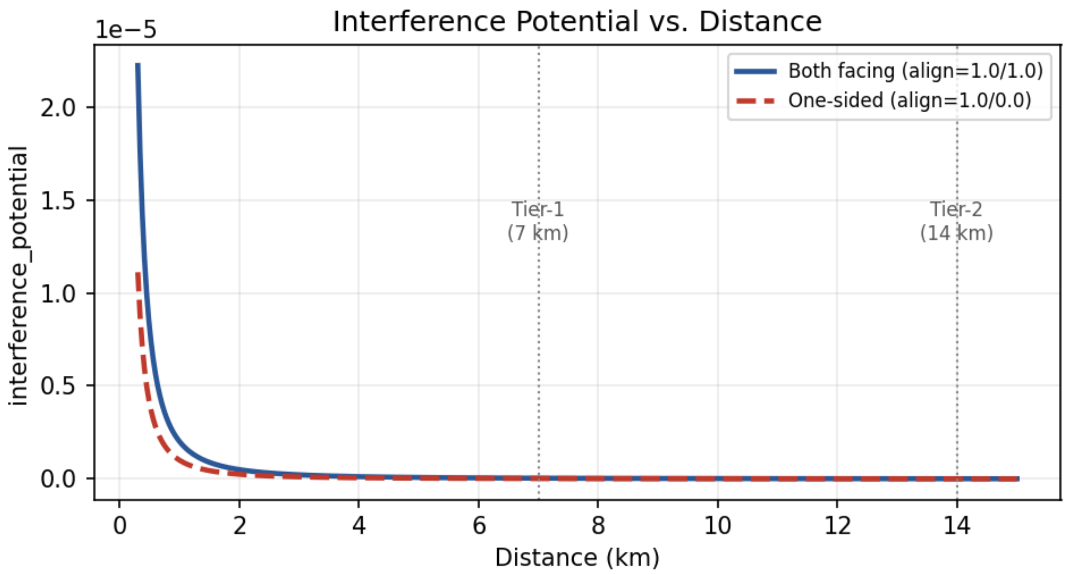

Fig. 3.3: Interference potential vs. distance for a fully-facing pair (both alignments = 1.0) and a one-sided pair (only one antenna facing). Tier-1 (7 km) and Tier-2 (14 km) boundaries are marked for reference.

Why not a binary cone?

A hard beamwidth cutoff (e.g. "within 60° = facing, beyond = not") creates a discontinuity with no physical basis: a neighbor at 59° and one at 61° would be treated as completely different cases despite nearly identical radiated energy in that direction. The cosine model in Equation (3.4) degrades smoothly instead. It is also deliberately parameter-free — no beamwidth or threshold constant had to be chosen or tuned, which matters because `cell_radius` in the available dataset is a uniform placeholder rather than per-site RF data, so any tuned cutoff would have been arbitrary rather than derived.

3.3.4 Stage 4: Constraint Generation

`constraints.py` answers exactly one question: *which PCI and RSI values are legal for a given sector, right now?* It never mutates assignments; it only computes forbidden and preferred sets from the current state.

Ring adjacency. Co-site sectors are ordered by azimuth and treated as a ring; `ring_adjacent_ids()` returns each sector's two immediate neighbors in that ring. This single helper backs both the Mod3 rule below and the co-site Mod4 rule in Section 3.4.1.

Site-size-aware Mod3 rule. Because only 3 Mod3 values exist, the rule strictness depends on how many sectors share a site:

Site size	**Mod3 rule**
1–3 co-located sectors	Every pair must differ (3 values are always sufficient for ≤ 3 sectors).
4–5 co-located sectors	Ring-adjacent pairs must differ; all 3 Mod3 values should appear on the site; any unavoidable reuse falls only on non-adjacent pairs.

Tab. 3.5: Mod3 rule by site size

Figure 3.4 illustrates the ring for a representative 4-sector site.

Measured site-size distribution. Figure 3.5 shows the distribution across all 725 sites in the dataset.

This distribution is the direct explanation for the soft-rule counts reported in Section 3.9: $431 + 2 = 433$ sites have 4 or 5 co-located sectors, and the measured Mod3 non-adjacent reuse count is exactly 433 sites — confirming the rule implementation behaves exactly as the site geometry predicts, not as an approximation of it.

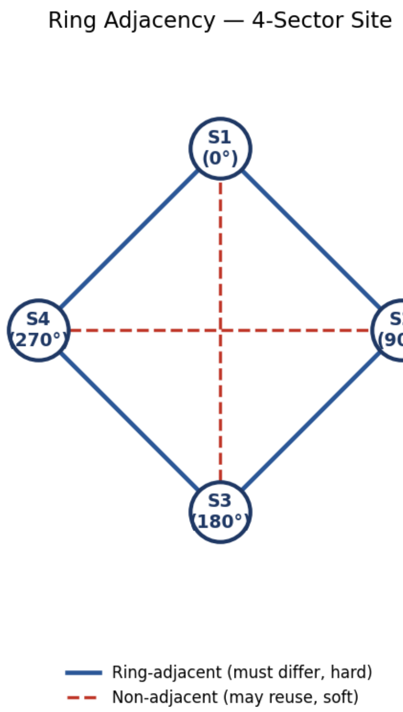

Fig. 3.4: Ring adjacency for a 4-sector site at 0°/90°/180°/270°. Ring pairs (solid) must differ under the hard rule; the two non-adjacent diagonal pairs (dashed) may reuse a Mod3/Mod4 value if the pigeonhole principle forces it.

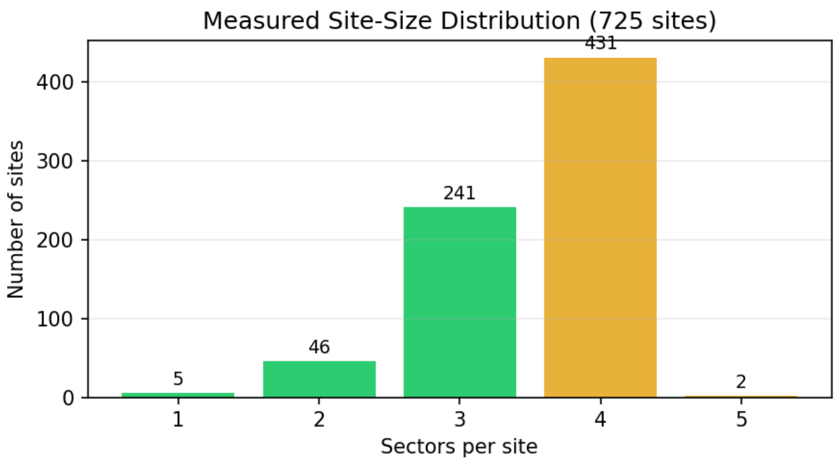

Fig. 3.5: Measured site-size distribution, full dataset. 431 + 2 = 433 sites fall under the 4–5-sector ring-adjacent-only rule.

3.4 Stage 5: The Assignment Engine

`assignment_engine.py` runs three sub-stages in strict order: Mod4, then PCI, then RSI. Every hard rule from Stage 4 is enforced without exception; the only freedom the engine has is in *which* legal value to pick, and in which *order* sectors are processed.

3.4.1 Sub-stage 5.1: Mod4 Assignment

Mod4 is planned as a **weighted graph-coloring problem**. Nodes are sectors; edges connect tier-1 inter-site pairs with nonzero `interference_potential` (Equation (3.5)); edge weight is that potential. Co-site pairs are excluded from this graph and handled by a separate co-site rule.

Co-site Mod4 rule. Analogous to the Mod3 rule but with only 4 values available:

- Sites with ≤ 4 sectors: every sibling pair must have a different Mod4.
- Sites with 5 sectors: only ring-adjacent siblings must differ (full pairwise separation is mathematically impossible: 5 sectors into 4 colors).

Step 1 — DSATUR initial coloring. The engine colors sectors using the degree-of-saturation (DSATUR) heuristic, adapted to a weighted objective.

Algorithm 1: Weighted DSATUR coloring for Mod4

lg:pi-dsatur **Input:** sectors to plan, interference edges, co-site forbidden-color function
Output: mod4_of: mapping from sector id to $\{0, 1, 2, 3\}$
Initialize `uncolored` $\leftarrow$ all sectors to plan;
while *`uncolored` is not empty* **do**
  Select $s^* = \arg\max_{s \in \texttt{uncolored}}$ (number of distinct colors already used by s's neighbors, tie-broken by total incident edge weight, then by sector id);
  Compute `forbidden` $\leftarrow$ co-site forbidden colors for s^*;
  `candidates` $\leftarrow \{0, 1, 2, 3\} \setminus$ `forbidden` (or all 4, if that is empty);
  Choose $c^* = \arg\min_{c \in \texttt{candidates}}$ (weighted same-color clash if s^* took color c), tie-broken deterministically by an MD5 hash of (s^*, c);
  Assign `mod4_of`$[s^*] \leftarrow c^*$; remove s^* from `uncolored`;

Step 2 — Local search repair. DSATUR alone does not reach a local optimum for a weighted objective, so a repair pass follows.

Algorithm 2: Weighted local search repair (recolor + swap)

lg:pi-repair **Input:** initial mod4_of, interference edges, co-site legality rules, time budget (60 s)

Output: repaired mod4_of

repeat

  `clashing` ← sectors with nonzero same-color weight, sorted worst-first;

  foreach $s \in \mathtt{clashing}$ **do**

    Evaluate every legal **recolor** of s to another color;

    Evaluate every legal **swap** of s's color with a co-site sibling's color;

  Apply the single move (recolor or swap) with the most negative total-weight delta;

until *no improving move exists* or *time budget exceeded*;

Termination guarantee

Total interference weight is bounded below by zero and *strictly* decreases every round that a move is applied, so the repair loop is guaranteed to terminate at a genuine local optimum. The 60 s wall-clock budget in **??** is a safety net for exceptionally dense pockets, not the expected termination path.

3.4.2 Sub-stage 5.2: PCI Assignment

With Mod4 fixed, `assign_pci_rsi()` processes sectors **site-by-site in azimuth (ring) order, hardest sites first** (ranked by the maximum tier-2 degree of any sector on the site), so that Mod3 site-coverage state can be tracked incrementally as siblings are placed one after another. For each sector, the candidate PCI pool is filtered through the hard rules — no tier-1 collision, no tier-2 confusion, not in a forbidden Mod3 group — and then narrowed by the four-layer preference funnel in Table 3.6.

Layer	Preference
1 (ideal)	Matches the sector's already-assigned Mod4 *and* fills a Mod3 value still missing on the site
2	Matches assigned Mod4 only
3	Fills a missing Mod3 value only
4 (fallback)	Any hard-legal candidate

Tab. 3.6: PCI layered preference funnel

Within whichever layer yields a non-empty pool, the PCI with the lowest global usage count is preferred; ties are broken with a deterministic MD5 hash of (sector id, salt) so that re-running the pipeline on unchanged input always reproduces *exactly* the same plan

— an explicit design requirement, verified empirically (Section 3.10) by running the full pipeline twice and diffing the outputs.

3.4.3 Sub-stage 5.3: RSI Assignment

RSI has no per-site structural rule — only global reuse-distance pressure — so sectors are processed **hardest-first by tier-2 neighbor count** instead of by site/azimuth order. Candidates exclude every RSI already used by a tier-2 neighbor. If that leaves nothing (a structurally dense pocket), the engine falls back to the least-conflicted RSI across the full range and logs a warning rather than raising an error, since exhaustive tier-2 exclusion failing indicates network density, not a bug. Ties are broken by least global usage, then the same deterministic hash used for PCI.

Why RSI reuse could be promoted to a hard rule

A degree-bound proof over the measured tier-2 neighbor counts (Table 3.4) confirmed that even the worst-case 14 km neighborhood — 344 sectors, the maximum observed — still leaves more than 480 of the 831 available RSI values free. This headroom is what justifies treating RSI Tier-2 reuse as a *hard* constraint rather than a soft one: the network is never so dense that zero-reuse RSI assignment becomes infeasible.

3.5 Stage 6: Independent Validation

`validator.py` recomputes every hard rule **completely independently** of the planner — it never reads planner-internal state, only the final `{pci, rsi, mod4}` assignments, and it re-derives Mod3/Mod4 from PCI itself rather than trusting whatever the engine wrote. It reuses `facing.py` from Stage 3, so the planner and the validator can never disagree about geometry. Every metric is reported at three granularities — by pair, by sector, and by site — and the overall `pass` flag is `True` only when *every* hard-rule count is exactly zero.

Rule	**Hard/Soft**	**Detection method**
PCI Collision	Hard	Any tier-1 pair sharing PCI
PCI Confusion	Hard	Any tier-2 pair sharing PCI
RSI Reuse	Hard	Any tier-2 pair sharing RSI
Mod3 Adjacent	Hard	Ring-adjacent co-site pair sharing Mod3
Mod3 Rule Violations	Hard	4–5 sector site not using all 3 Mod3 values
Mod4 Intra-site	Hard	Ring-adjacent co-site pair sharing Mod4
Mod3 Non-Adj Reuse	Soft	Non-adjacent co-site pair sharing Mod3
Mod4 Non-Adj Intra-site	Soft	Non-adjacent co-site pair sharing Mod4
Mod4 Inter-site	Soft	Sector's nearest facing inter-site neighbor shares Mod4

Tab. 3.7: Rules checked by the independent validator

The Mod4 Inter-site check deserves special mention: it is **per-sector and directed**, matching the definition used by the reference clash sheet this engine was built against. For sector S, find the nearest tier-1 neighbor on a different site that S's antenna actually points at ($\texttt{alignment}(S, N) > 0$), and count one clash if $\text{Mod4}(S) = \text{Mod4}(N)$. Its maximum possible value therefore equals the total sector count (2,554), not the pair count.

3.6 Stage 7: Export

`exporter.py` writes `PCI/RSI/Mod4` back into the source workbook's `Planning` sheet by matching rows on `site_sector_ID` — never by row position, since row order is not guaranteed to match assignment order — and appends a `KPI_Report` sheet generated directly from the validator's report dictionary. There is no separate calculation path for the exported numbers, so the numbers a reviewer sees in the spreadsheet and the numbers the validator computed can never drift apart.

3.7 Planning Modes

	Bulk planning (`bulk_plan.py`)	**Single-site planning (`add_site.py`)**
Scope	Entire network, region by region	One new site's sectors only
Locked assignments	None (full reset)	Every other already-planned sector in the site's region
Engine call	`assign_all()` per region	`assign_all()` for the target site's sectors
Measured runtime	53.36 s total, 4 regions (Section 3.10)	Well under 5 s target; scoped to one site's local neighborhood

Tab. 3.8: Bulk planning vs. single-site planning

Both callers share the *exact same* `assign_all()` function; the only difference is what is pre-populated in `existing_assignments`. `add_site.py` additionally filters the neighbor graph and site grouping down to the new site's region only, using the same `region_of()` prefix rule that `bulk_plan.py` uses.

3.8 Hard Rules Reference

Rule	Distance/scope	Why it matters
PCI Collision	7 km (tier-1)	Two sectors this close sharing a PCI are indistinguishable to a UE.
PCI Confusion	14 km (tier-2)	Sharing a PCI this far apart still confuses which physical cell a UE actually saw, disrupting handover and measurement reporting.
RSI Reuse	14 km (tier-2)	Shared PRACH root sequence causes random-access preamble collisions between UEs on different cells.
Mod3 Adjacent	Co-site, ring-adjacent	Ring-adjacent sectors sharing Mod3 share a PSS group, degrading synchronization-signal separation at the sector boundary.
Mod3 Rule (strict)	Co-site, 4–5 sectors	All 3 Mod3 values must appear on the site; unavoidable reuse must land only on non-adjacent pairs.
Mod4 Intra-site	Co-site, ring-adjacent	Ring-adjacent sectors sharing Mod4 share SRS/DMRS resource allocation, degrading uplink channel estimation at the boundary.

Tab. 3.9: Hard rules enforced by the planning engine (all must equal zero)

Measured result

All six hard-rule checks above returned exactly zero across all 2,554 sectors and 725 sites in the verified full-reset run — **PASS**.

3.9 Soft Rules Reference

Each soft rule exists because the network's own density creates a pigeonhole constraint — not because the planner failed to try. Table 3.10 reports measured values alongside the structural expectation each should match.

Rule	Structural cause	Measured	Structural expectation
Mod3 Non-Adjacent Reuse (sites)	3 Mod3 values, 4–5 sectors on 433 sites	433	433 (= count of 4–5-sector sites)
Mod4 Non-Adjacent Intra-site (sites)	4 Mod4 values, 5 sectors on 2 sites	2	2 (= count of 5-sector sites)
Mod4 Inter-site (sectors)	4 Mod4 values against ∼100 avg. tier-1 neighbors/sector	517 (20.2%)	No exact closed-form floor; minimized by **????**, bounded below by network density

Tab. 3.10: Soft rules, structural cause, and measured results

The first two soft metrics match their structural expectation *exactly*, confirming the site-size-aware rule implementation of Section 3.3.4 behaves as designed. The third has no simple closed-form floor: it is the outcome of the DSATUR-plus-local-search process in Section 3.4.1, minimized as far as that local search can reach rather than eliminated outright.

Mod4 inter-site is a structural ceiling — not an algorithm limitation

Single-sector tabu-style local search plateaus almost immediately at ∼7 km neighborhood density: with ∼100 tier-1 neighbors competing for only 4 colors, there are simply not enough degrees of freedom for any single-sector move to eliminate every clash. Further reduction requires *coordinated multi-sector* moves — changing several sectors' colors together in a way that no single-sector move can discover on its own. This is an identified direction for future work rather than a defect in the current repair pass.

3.10 Measured Results and Performance

Figure 3.6 summarizes every measured KPI from a single full-reset run of the pipeline against the live 2,554-sector, 725-site dataset.

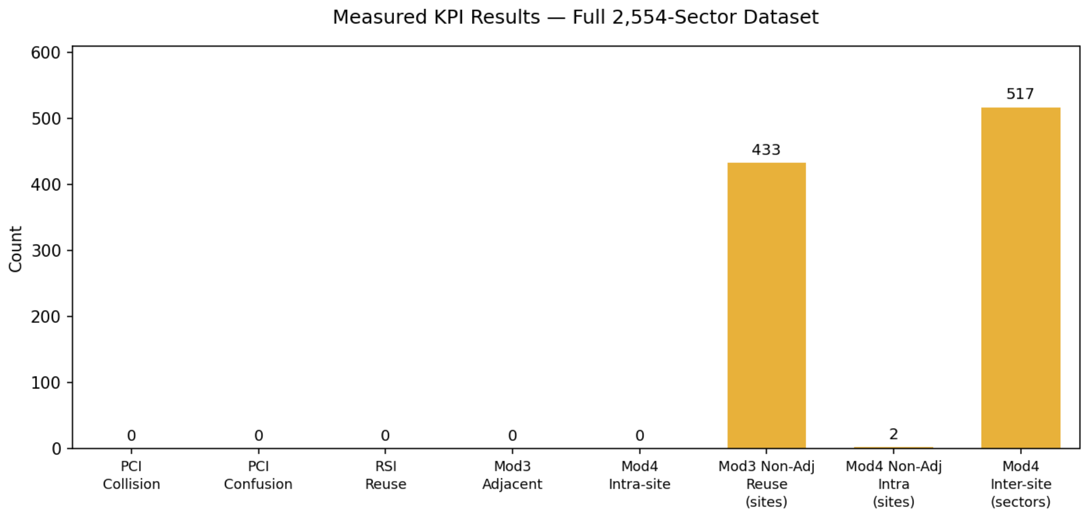

Fig. 3.6: Measured KPI results, full dataset, single full-reset run. Green bars are hard rules (all zero); amber bars are soft, structurally-expected counts.

Metric	Measured value
Sectors planned	2,554
Sites	725
Regions	4 (ALX, SIN, UPP, DEL)
PCI Collision (hard)	0
PCI Confusion (hard)	0
RSI Reuse (hard)	0
Mod3 Adjacent (hard)	0
Mod3 Rule Violations (hard)	0
Mod4 Intra-site, ring-adjacent (hard)	0
Mod3 Non-Adjacent Reuse (soft, sites)	433 of 433 eligible 4–5-sector sites
Mod4 Non-Adjacent Intra-site (soft, sites)	2 of 2 eligible 5-sector sites
Mod4 Inter-site (soft, sectors)	517 of 2,554 (20.2%)
`bulk_plan()` runtime	53.36 s total
`validate()` runtime	0.42 s
Overall result	**PASS**

Tab. 3.11: Full measured results summary

3.10.1 Runtime Benchmarks

Figure 3.7 and Table 3.12 break the total planning time down by region. Regions are processed largest-first, which is why Alexandria and Sinai dominate total wall-clock time:

both the interference graph and the local-search repair scale with each region's own density, not the global dataset size, consistent with the region-independence property of Section 3.2.

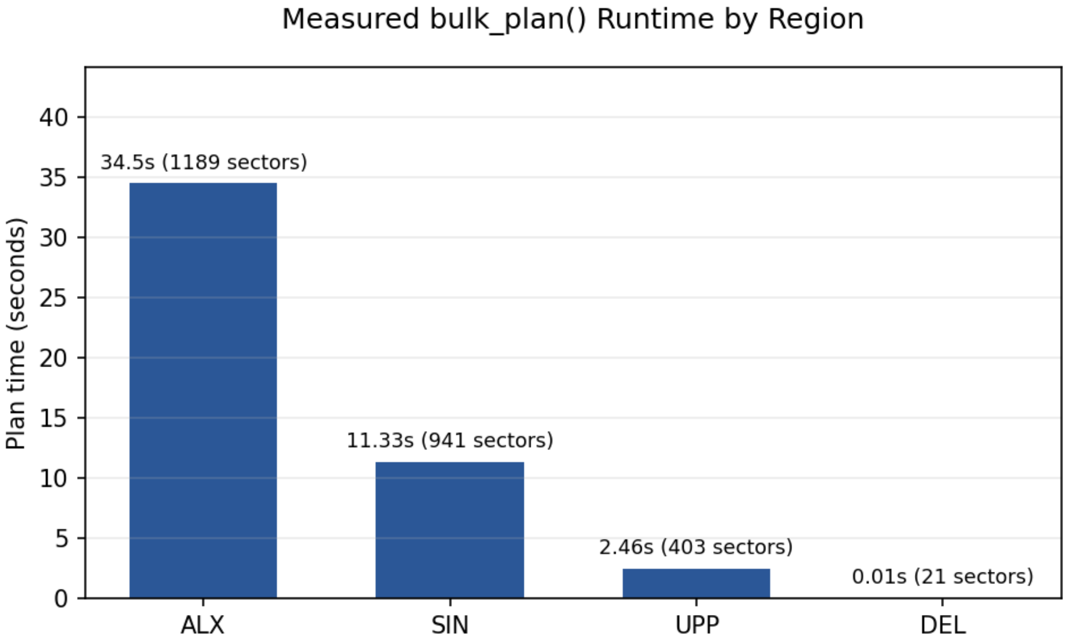

Fig. 3.7: `bulk_plan()` wall-clock time per region, measured on the full dataset.

Region	**Sectors**	**Sites**	**Plan time (s)**
ALX	1,189	338	34.50
SIN	941	266	11.33
UPP	403	115	2.46
DEL	21	6	0.01
Total	**2,554**	**725**	**53.36**

Tab. 3.12: Measured per-region runtime

The independent `validate()` call on the resulting assignments completed in 0.42 s — a deliberate design outcome, since a validator that is itself slow would discourage running it after every change.

Determinism — verified empirically

The full pipeline was run independently more than once on the unchanged raw dataset. Every PCI, RSI, and Mod4 value matched exactly across runs — zero mismatches. This determinism follows directly from the MD5-hash tie-breaking described in Sections 3.4 and 3.4.1: given identical input, every ordering and every tie-break resolves identically, so the plan is fully reproducible rather than dependent on iteration order or randomness.

3.11 Chapter Summary

The Planning Intelligence engine assigns PCI, RSI, and Mod4/Mod3 to every sector of a 2,554-sector, 725-site live network through a seven-stage, flat-function pipeline. Mod4 is solved as a weighted graph-coloring problem (DSATUR plus guaranteed-terminating local search repair); PCI is chosen through a four-layer preference funnel that respects Mod4 and Mod3 simultaneously; RSI is assigned through a global tier-2 exclusion scheme justified by a degree-bound feasibility proof. An independent validator, sharing only the geometry module with the planner, confirmed **zero hard-rule violations** across the entire dataset, and the two soft-rule metrics that have exact structural expectations matched them precisely. The remaining soft metric — Mod4 inter-site clashes — represents a genuine, density-driven structural ceiling for single-sector optimization, and is the basis for the multi-sector coordinated-move extension proposed as future work in **??**.

CHAPTER 4

Integration and Platform